

Chapter 17 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (15 points)

1-5: EBDAC

6-10: AABDC

11-15: EBDED

Part B short question (15 points)

16. (3 points)

- (a) Stay the same.
 (b) The concentration of B(aq) increases.
 (c) pH of the solution decreases.

17. (6 points)

(a)

$$\text{pH} = \text{p}K_a + \log \frac{[\text{base}]}{[\text{acid}]}$$

$$[\text{base}] = [\text{sodium acetate}] = \frac{0.13 \text{ mol}}{1.00 \text{ L}} = 0.13 \text{ M}$$

$$[\text{acid}] = [\text{acetic acid}] = \frac{0.10 \text{ mol}}{1.00 \text{ L}} = 0.10 \text{ M}$$

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log \frac{0.13}{0.10} = 4.7447 + 0.11394 = 4.8587 = 4.86$$

(b)

$$[\text{base}] = [\text{sodium acetate}] = \frac{0.13 + 0.02}{1.00 \text{ L}} = 0.15 \text{ M}$$

$$[\text{acid}] = [\text{acetic acid}] = \frac{0.10 - 0.02}{1.00} = 0.08 \text{ M}$$

	CH ₃ COOH(aq)	⇌	CH ₃ COO ⁻ (aq)	+	H ⁺ (l)
In. Con.	0.08 M		0.15 M		0
Ch. Con.	-x M		(0.15+x) M		x M
Eq. Con.	(0.08-x) M		(0.15+x) M		x M

$$K_a = \frac{x(0.15 + x)}{0.08 - x} = 1.8 \times 10^{-5}$$

$$[\text{H}^+] = x \approx 9.6 \times 10^{-6}$$

$$\text{pH} = -\log[\text{H}^+] = 5.0177 = 5.02$$

OR

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log \frac{0.15}{0.08} = 4.7447 + 0.2730 = 5.0177 = 5.02$$

(c)

$$[\text{base}] = [\text{sodium acetate}] = \frac{0.13 - 0.02}{1.00 \text{ L}} = 0.11 \text{ M}$$

$$[\text{acid}] = [\text{acetic acid}] = \frac{0.10 + 0.02}{1.00} = 0.12 \text{ M}$$

	CH ₃ COOH(aq)	⇌	CH ₃ COO ⁻ (aq)	+	H ⁺ (l)
In. Con.	0.12 M		0.11 M		0
Ch. Con.	-x M		(0.11+x) M		x M
Eq. Con.	(0.12-x) M		(0.11+x) M		x M

$$K_a = \frac{x(0.11 + x)}{0.12 - x} = 1.8 \times 10^{-5}$$

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$$[\text{H}^+] = x \approx 1.964 \times 10^{-5}$$

$$\text{pH} = -\log[\text{H}^+] = 4.7069 = 4.71$$

$$\text{pH} = -\log(1.8 \times 10^{-5}) + \log \frac{0.11}{0.12} = 4.7447 - 0.03779 = 4.70691 = 4.71$$

18. (6 points)

(a)

$$K_{sp} = [\text{Ag}^+][\text{I}^-] = 8.3 \times 10^{-17}$$

$$[\text{Ag}^+] = [\text{I}^-] = 9.11 \times 10^{-9} = 9.1 \times 10^{-9} \text{ M}$$

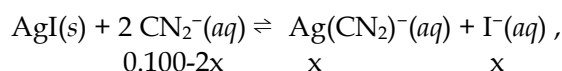
(b) calculate the equilibrium constant for the reaction $\text{AgI}(s) + 2 \text{CN}^-(aq) \rightleftharpoons \text{Ag}(\text{CN})_2^-(aq) + \text{I}^-(aq)$,

$$K = K_{sp} \times K_f = 8.3 \times 10^{-17} \times 1.0 \times 10^{21} = 8.3 \times 10^4$$

(c)

K is much greater than one for the reaction of $\text{AgI}(s)$ with CN^- . This means that the reaction goes to completion. For a $\text{AgI}(s)$ in 0.100 M NaCN solution, CN^- is the limiting reactant. Two moles of CN^- react with one mole of AgI , so the solubility of AgI in 0.100 M NaCN is $(0.100/2) = 0.0500 \text{ M}$.

OR



$$K = \frac{[\text{Ag}(\text{CN}_2)^-][\text{I}^-]}{[\text{CN}_2^-]^2} = \frac{x^2}{(0.100 - 2x)^2} = 8.3 \times 10^4$$

$$[\text{I}^-] = x = 0.05 \text{ M}$$